Instructions for exam

Hello!

If a question is unclear, please write down your assumptions about the question when you are answering it.

The exam has a total of 40 points. You need 21 points to pass the exam. The grade is on the scale U/G.

Answers must be written in English. Dictionaries may be used.

It is possible to get partial credit for the questions that aren't multiple choice.

A list of tips to support you with the exam:

- Answer all questions. A bad answer never gives less points than no answer.
- Read the question again after you have written the answer. Verify that you have answered the question asked. Verify that you answered all parts.
- Don't forget to give an example if that is requested.
- When a question asks you to compare two things A and B, make sure to highlight their differences. It is better to use tables than paragraphs of text to compare.

I likely will not be able to make it to the exam hall today due to scheduling conflicts but I wish you all the best!

1 Multiple Choice

Each question in the multiple choice section is worth 1 point. There is only one expected best answer for each question.

What is the primary goal of risk management in software engineering?

- To identify, assess, and mitigate potential risks
- To allocate extra budget for unexpected failures
- To remove all possible risks from the project
- To ensure the software is developed without any defects

Which technique is used to gather requirements from stakeholders?

- Requirements elicitation
- Debugging
- Code inspection
- Continuous deployment

Which of the following software development models follows a linear, sequential approach?

- Incremental
- Agile
- Waterfall
- Spiral

What is the main difference between black-box testing and glass-box testing?

- Black-box testing focuses on internal code structure, while glass-box testing examines system functionality
- Black-box testing only applies to unit tests, while glass-box testing applies to integration tests
- Black-box testing does not require access to the code, while glass-box testing requires knowledge of the code structure
- Black-box testing uses source code analysis, while glass-box testing tests from a user perspective

How does Test-Driven Development (TDD) differ from traditional development approaches?

- TDD eliminates the need for validation and verification
- TDD writes tests after implementing the functionality
- TDD focuses only on UI testing
- TDD requires writing tests before writing the actual code

Which of the following best describes continuous integration?

- A development practice where code changes are frequently integrated into the main branch and tested
- A testing method used for finding performance issues
- A method of organizing teams in Agile development
- A version control approach that delays integration until the end of a project

In peer review, how does the process typically work?

- One programmer presents their work while another follows along in real time
- The reviewer makes modifications directly to the programmer’s code
- The entire team meets in a formal session to discuss previously reviewed code
- The reviewer inspects a portion of source code and provides feedback

What is the primary purpose of code reviews?

- To completely eliminate all bugs before release
- To reduce defect rates and improve maintainability
- To allow managers to evaluate programmers' performance
- To rewrite code from scratch

Which of the following is a key element of Agile planning?

- Long-term fixed milestones
- Iteration planning and release planning
- One-time budgeting at the start of the project
- Fixed project scope

In project planning, what does a Gantt Chart (Activity bar chart) primarily help visualize?

- The project schedule and task dependencies
- The functional and non-functional requirements
- The project's overall quality
- The relationship between stakeholders

What is one key benefit of having an explicit architecture?

- Reduces the need for stakeholder communication
- Ensures the system design never needs modifications
- Eliminates the need for software documentation
- Helps analyze whether the system meets non-functional requirements

What is a challenge with requirements elicitation?

- There is no need to validate requirements after they are collected
- Requirements remain constant throughout the development process
- Stakeholders may not fully understand or articulate their needs
- Stakeholders always agree on requirements

What is the COCOMO model used for?

- Estimating the cost of software development
- Evaluating software security risks
- Managing team roles in software projects
- Creating Agile sprint backlogs

Totalpoäng: 13

2 Requirements

What is a difference between user requirements and system requirements (1pt)?

Fill in your answer here

A local pharmacy has decided to start an online platform. They have hired your company to create the software system.

Identify three stakeholders and an example of a functional requirement they would be interested in. (3 pt)

Fill in your answer here

The client has stated the following, "I want the customers to be able to find their medication easily."

From this statement, write an example of the following **two** things. You are allowed to make assumptions if you don't have enough information.

(a) - a user story (1 pt)

Fill in your answer here

(b) - a non-functional requirement (1 pt)

Fill in your answer here

Totalpoäng: 6

3 Validation and Verification

In the context of software engineering, differentiate between validation and verification. (2 pt)

Fill in your answer here

Describe how validation can be assessed by explaining two techniques that can be used to ensure requirements correctness. (2 pt)

Fill in your answer here

Totalpoäng: 4

4 Code Inspection

A company is debating between using **pair programming** and **pre-checking** for their code inspection process. Name three factors should they consider when choosing between these two approaches? (3 pts)

Fill in your answer here

Totalpoäng: 3

5 Architecture

Consider a hospital management system that uses an **MVC architecture**. The management system tracks things such as inventory, patients, employee and patient information. Identify the likely responsibilities of the **Model, View, and Controller** components in a possible use case of the software. This use case does not need to be complex. A sketch may be useful. (5 pt)

Fill in your answer here

Teckenf…

Ord: 0

Totalpoäng: 5

6 Legacy Systems

A legacy system in a hospital is still in use but has no formal documentation. What are two metrics that can be used to assess whether it should be maintained or replaced? A diagram may be useful. Explain your answer. (4 pt)

Fill in your answer here

Teckenf…

Ord: 0

Totalpoäng: 4

7 Project Management

Here is an activity table:

Task	Effort (person-days)	Duration (days)	Dependencies
T1	6	3	
T2	4	4	T1 (M1)
T3	1	1	T1 (M1)
T4	4	8	T3 (M2)

Sketch the activity bar chart (3 pt). What are two pieces of information you need in order to create a staff allocation chart (2 pt)?

Fill in your answer here

Teckenf…

Ord: 0

Totalpoäng: 5